Reporting discoveries from LLM-driven evolutionary search

A practitioner's rule, and the evidence behind it

Banana Labs · 2026-08-18 · assessment of the 2026-08-18 direction · full report: `transfer-sn/DIRECTION_ASSESSMENT_2026-08-18.html`

If you read nothing else. Use a strong ensemble, run it 5 to 10 times with the same settings, check the *circuits* rather than the commentary, and confirm by actually applying the transformation to the inputs. The model's own write-up of what it discovered carries no evidential weight: the mid-tier ensemble claimed the symmetry in 5 runs out of 5 and built it in 1.

WHAT TO NOTICE

The model's write-up tells you nothing about whether it found something. Mid-tier ensembles wrote confident symmetry notes in *every* run, 5/5, and built the target structure in one. A researcher reading only the patch notes would have concluded five times over that the symmetry was found.

A high score does not merely fail, it points the wrong way. The two best-scoring weak runs both had the wrong structure; the correct structure appeared at ranks 3 and 4. The natural move, take the winner and study it, is the move that misleads you.

Wild swings between identical reruns mean the instrument is not working. Ten byte-identical weak runs spanned 0.00 to 0.75, and one never improved on the starting circuit at all.

WHAT TO DO

1. **Run it 5 times, ideally 10, same settings.** Once is not an experiment.
2. **Pull the actual circuit from each run and check whether the same structural pattern recurs.** Does one shared parameter drive the same gate across all qubits? Not "does the model mention it" but "does the artifact contain it". Ours: 3/3 strong, 1/5 mid, 2/10 weak. This is the test.
3. **Then test the behaviour directly.** If you think you found "relabelling the qubits does not matter", relabel them and check the output is unchanged. No judgement call about which pattern to look for. Trust this one most.
4. **Fix what counts as a hit before looking at results.** We chose ours knowing the answer: fine for a study, not for real work.
5. **If only one model ever produces the structure, be suspicious.** One model found it 18/20 cold starts against 1/21 for the other two. That may be about the model, not your problem.

THE ONE THING THIS CANNOT YET TELL YOU

If reruns *disagree*, that means "this resolved nothing", not "there is no symmetry". The two are indistinguishable from where you sit. Weak and mid gave inconsistent results on a problem that genuinely does have the symmetry, purely because they were not good enough to find it. We know a *strong* ensemble correctly returns nothing when there is nothing (0/3 on a dataset built with no symmetry). What weak and mid do in that situation is the experiment still pending.

EVIDENCE

	weak	mid	frontier
runs (n)	10	8	3
mean score	0.373	0.365	**0.988**
sd / CV	0.239 / 0.64	0.222 / 0.61	**0.054 / 0.05**
named any symmetry	8.5%	**81.2%**	40.8%
named *permutation*	0%	**0%**	30.6%
named *mirror*	2.1%	**75.0%**	2.1%
built the S_8 orbit	27.7%	**0%**	83.3%
motif in best program	2/10	1/5	**3/3**

Negative control (2026-08-18): three frontier runs on a dataset identical except that the label rule was changed to destroy the symmetry. Best programs kept 16, 24 and 20 free parameters against the seed's 24, with **zero** tied families, versus 4, 2 and 5 parameters and 3/3 on the real task. Fisher $p = 0.10$, pre-registered as suggestive. The null runs *did* propose tied circuits early; the fitness function rejected them.

CORRECTING THE FRAMING

Not hallucination. Matched claim to matched structure, mid's follow-through on the symmetry *it* named is 35/36 = 97%. It named *mirror* symmetry (a real symmetry of the circuit layout, not of the task) in 75% of proposals and *permutation* symmetry in 0%, and built mirror faithfully. The defect is *hypothesis lock-in*, not honesty, and that is the stronger claim.

Verbal consistency across reruns fails as a rule. Mid satisfies it 5/5 and is wrong in 4. Only structural recurrence separates.

Guideline, not benchmark. weak vs mid is unresolvable ($d = 0.19$, $\sim$63 runs/arm) and weak outscores mid at a third the cost.

PRIOR ART, AND THE GAP

Stability selection (Meinshausen & Bühlmann 2010) is the statistical ancestor but perturbs the data, whereas here the randomness is LLM sampling. AlphaEvolve, TurboEvolve and QuantaAlpha report rerun spread, always as an error bar and never as an inferential quantity. arXiv:2605.20086 is the likeliest partial scoop and is known only from a search snippet; read it before drafting. Nothing found treats rerun variance as a diagnostic for whether a discovery is *real*, or validates such a criterion against a known hidden symmetry *plus* a matched negative control.

NEXT, BY VALUE PER DOLLAR

(1) Behavioural equivariance on the six best programs already extracted, $\sim$free, removes the protocol's most attackable point. (2) Weak and mid null arms, $\sim$\$9, closes the gap above. (3) Null replicates 3 $\rightarrow$ 10, $\sim$\$40, moves Fisher below 0.05. (4) A second symmetry class (`zc-su2`), fingerprint pre-registered, defends against "permutation invariance is over-represented in pretraining". Not yet scheduled.

Every number is quoted from an artifact in `transfer-sn/`: `VALIDATION_CRITERIA.md`, `SAY_VS_BUILD.md`, `EXP1_RESULTS.md`, `EXP1B_RESULTS.md`, `DISCOVERY_ATTRIBUTION.md`, `RELATED_WORK.md`, `BENCHMARK_PLAN.md`, `NULL_CONTROL_REPORT.html`, `launch_null.sh`, `launch_null_tiers.sh`, `onepager_symmetry/report.tex` and `onepager_symmetry/mirror_stats.json`. Threats to the result, stated in full in the HTML report: the structural fingerprint was chosen knowing the answer and a looser one collapses the separation; $p = 0.10$ at $n = 3$; ensemble members share pretraining so their agreement is not independent; the frontier effect may be one model (gpt-5.6-sol, 18/20 cold starts against 1/21, Fisher $p = 1.7 \times 10^{-8}$); a reliably failing searcher also has low variance. No claim rests on a run that had not completed at the time of writing.